

Gregory Lim

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Education

Singapore University of Technology and Design

Bachelor of Engineering in Computer Science

Singapore

Sep 2022 – May 2026

Experience

Open Source Contributor

Google DeepMind

- Solved 19-year-old open problem in mathematics by proving finite, simple, K_3 -free, induced- P_5 -free graphs are well-totally-dominated, and submitting verified Lean proof in Formal Conjectures under PR #4455.

AI Engineer Intern

Creatz3D

May 2025 – Sep 2025

Singapore

- Introduced industry-novel capability by developing proprietary GraphSAGE-style mesh thickening models.

Proficiencies

- **Languages.** C, CSS, HTML, JS, Python, Shell, TeX, and more.
- **Tools.** Git, Gradio, HF, LangChain, n8n, NumPy, pandas, PyTorch and more.

Projects

Vibe Decoding

github.com/glyaea/vibe-decoding

- Introduced auxiliary-model-free, single-hyperparameter, $O(Vd)$ -time next-token rule that generalises greedy decoding by aligning embeddings with their expectation.
- Characterised asymptotic runtime and greedy hyperparameter interval of rule.
- Outperformed macro F1 and MRR of Qwen 2.5 0.5B, Gemma 2 2B, and Llama 3.1 8B on greedy decoding by up to 10.03% on BoolQ, and up to 5.12% on NumerSense.

SUTDoko

sutdoko.vercel.app

- Built typo-resistant SUTD room name look-up tool by training language models using decoding-constrained character-level transformers for room name denoising task on handcrafted dataset.
- Made web inference smoother and over 2 times faster by making it cached, debounced, and incremental.
- Outperformed official SUTD Assistive Maps Telegram bot by 100% on subset of test set.

Recognition

- Information Systems Prize, Singapore University of Technology and Design & Singtel, 2024.
- Mathematics and Science Book Prize, Temasek Secondary School, 2017.